📄 PS-8 : 8. Jenkins a. Create a repository on GitHub and upload project files. b. Install and launch Jenkins (http://localhost:8080). c. Install required plugins (Git, GitHub, Pipeline). d. Create a new Pipeline job in Jenkins. e. Configure Pipeline script from SCM and add GitHub repository URL. f. Add a Jenkinsfile to the repository. g. Configure build trigger (Poll SCM or GitHub Webhook). h. Push code changes to GitHub. i. Verify that Jenkins automatically triggers the build

— FULL PROCESS (Windows)

🔹 1. Install Required Software

Install:

- Java (JDK 17+)
- Git
- Python

Check in Command Prompt:

java -version
git --version
python --version

🔹 2. Download and Run Jenkins

Download .war file from Jenkins run it on browser it will directly download

https://get.jenkins.io/war-stable/latest/jenkins.war

(save the above file (jenkins.war) path to a folder in C drive in Jenkins folder)

Move it to:

C:\Jenkins

Run Jenkins:

cd C:\Jenkins
java -jar jenkins.war --webroot=C:\Jenkins\temp

🔹 3. Unlock Jenkins

Open browser:

http://localhost:8080

Get password:

type "C:\Users\YOUR_USERNAME\.jenkins\secrets\initialAdminPassword"

Copy and Paste the password → Continue → Install suggested plugins → Create admin user

---

- 4. Create Project Using Git Bash

Open Git Bash

```
mkdir AI-CI-Project
cd AI-CI-Project

touch app.py test_app.py requirements.txt Jenkinsfile
```

- 5. Add Code (using Notepad)

Open files on git bash

notepad app.py

Paste:

```
def add(a,b):
  return a+b
```

---

notepad test_app.py

Paste:

```
import unittest
from app import add

class TestAdd(unittest.TestCase):
  def test_add(self):
    self.assertEqual(add(2,3),5)

if __name__ == "__main__":
  unittest.main()
```

---

notepad requirements.txt

Paste:

pytest

---

notepad Jenkinsfile

Paste:

```
pipeline {
  agent any

  stages {
    stage('Install Dependencies') {
      steps {
        bat 'pip3 install -r requirements.txt'
      }
    }

    stage('Run Tests') {
      steps {
        bat 'python -m unittest'
      }
    }
  }
}
```

- 6. Initialize Git and Commit on bash

```
git init
git add .
git commit -m "Initial CI project"
```

- 7. Create Repository on GitHub

Go to GitHub

- Create new repo: AI-CI-Project

🔹 8. Connect Local Project to GitHub

git remote add origin https://github.com/YOUR_USERNAME/AI-CI-Project.git
git branch -M main
git push -u origin main

🔹 9. Create Pipeline Job in Jenkins

Open Jenkins

👉 Click **New Item**

- Name: AI-CI-Pipeline
- Select: Pipeline

🔹 10. Configure Pipeline

Build Trigger:

✓ Select: **Poll SCM**

Enter:

H/2 * * * *

Pipeline Section:

Field	Value
Definition	Pipeline script from SCM
SCM	**Git**
Repository URL	**Your GitHub repo-> https://github.com/USER-NAME /AI-CIProject.git**
Branch	***/main**
Script Path	**Jenkinsfile**

- 11. Run First Build

Click:

Build Now

- 12. Check Output

Click:

Console Output

Expected:

Ran 1 test
OK
Finished: SUCCESS